



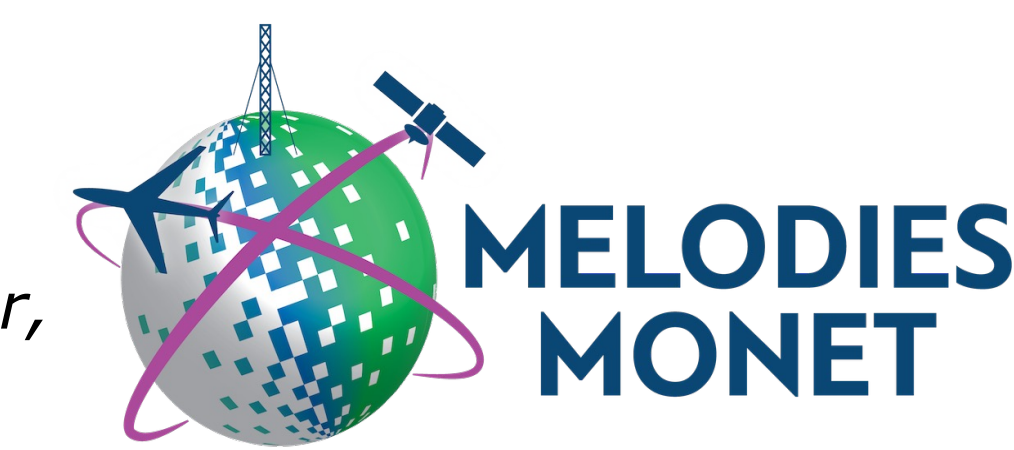
Developing an Open-Source Meteorological Evaluation Toolkit within MELODIES-MONET

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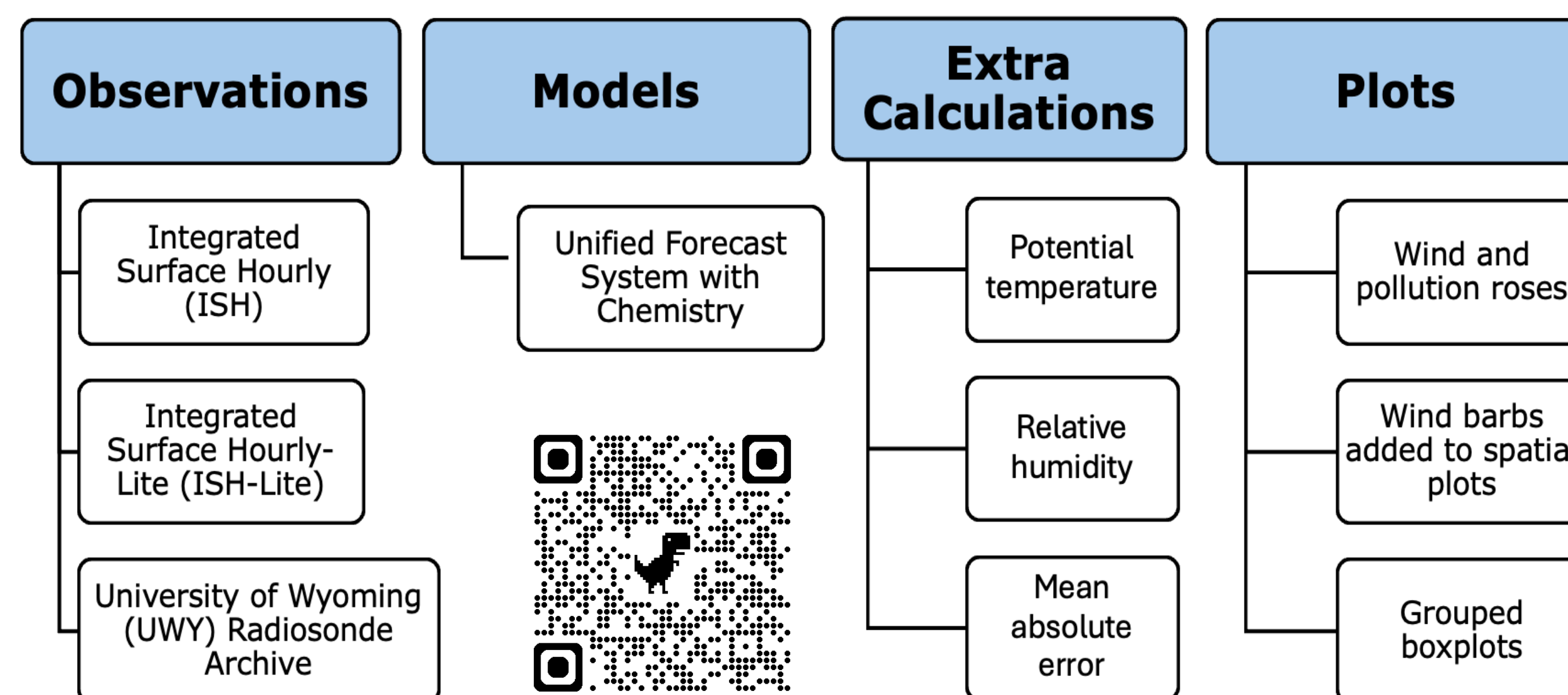
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AGU25
New Orleans, LA | 15–19 December 2025



i) Background

- MELODIES-MONET (**M**odel **E**va**L**uation using **O**bservations, **D**iagnos**I**c and **E**xperiments **S**oftware – **M**odel and **O**bservation **E**valuation **T**oolkit) is a Python-based model evaluation toolkit
- **Meteorological evaluation** capabilities within MELODIES-MONET **are limited**
- **This project develops an open-source meteorological toolkit** within MELODIES-MONET



Schematic overview of the meteorological evaluation capabilities MELODIES-MONET users can now leverage

ii) Meteorological Showcase (fig 1-6)

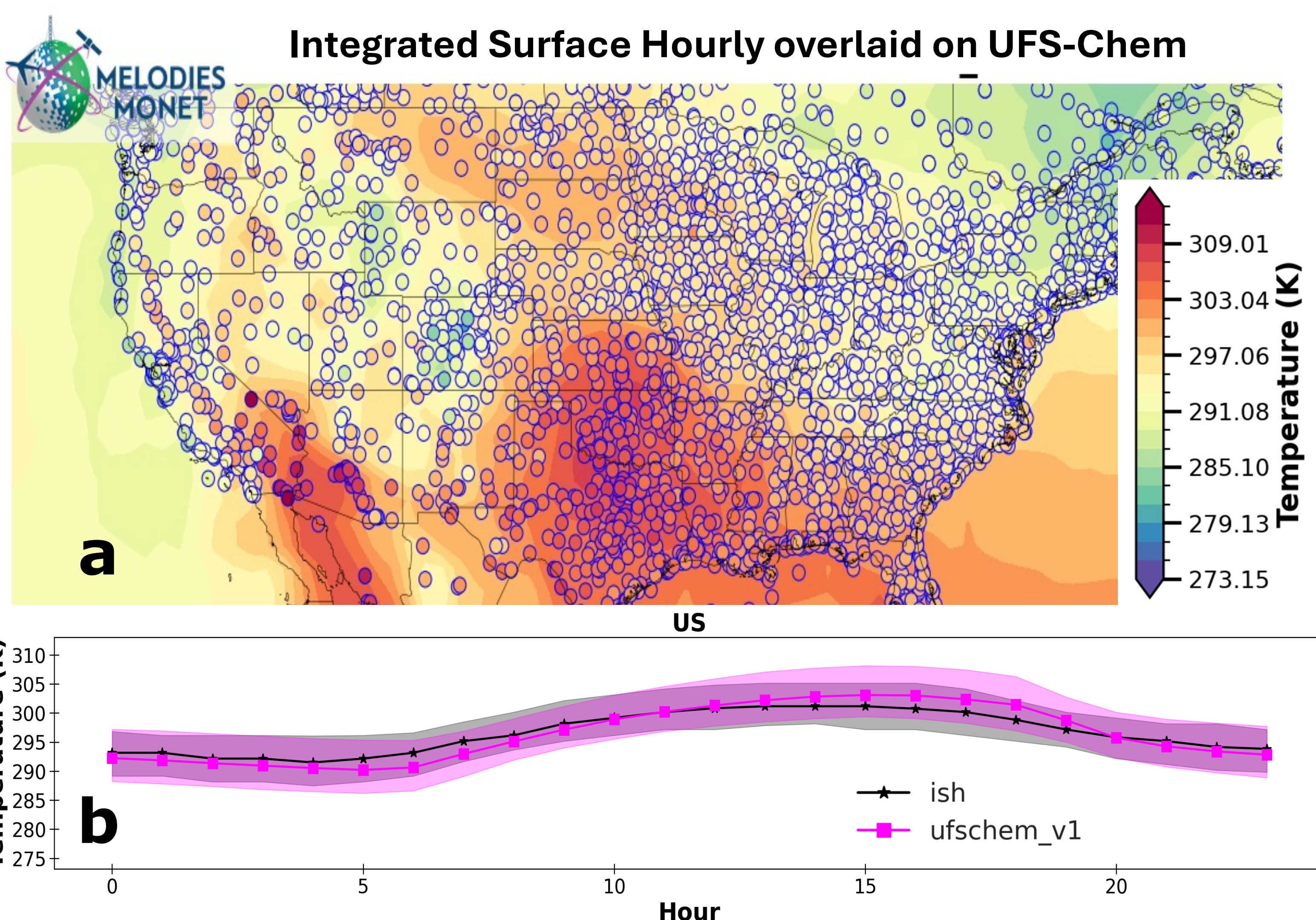


Fig. 1a & b: Users can conduct spatial and diurnal analyses of atmospheric variables such as temperature

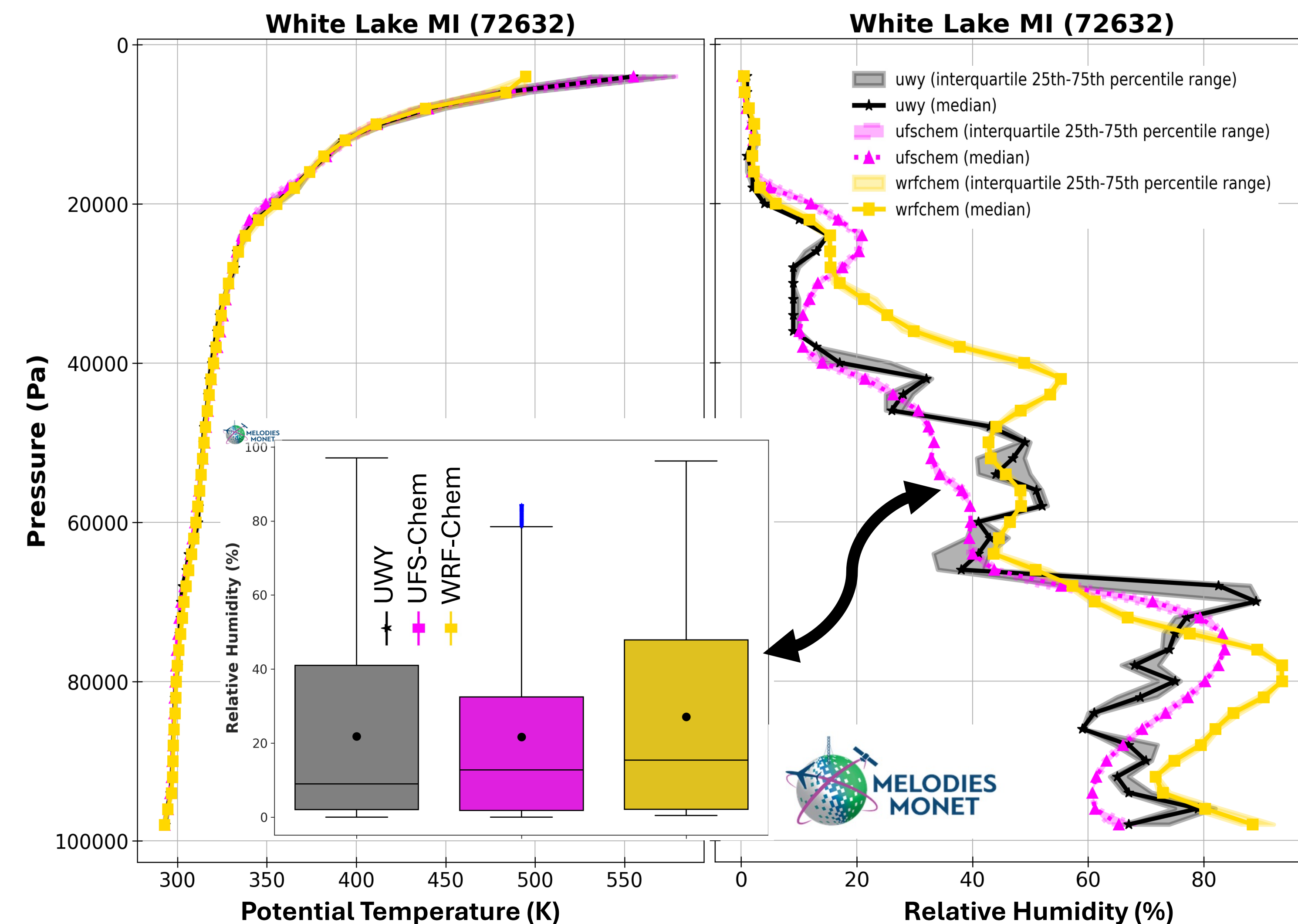


Fig. 2: Variables such as potential temperature and relative humidity can be directly calculated and plotted

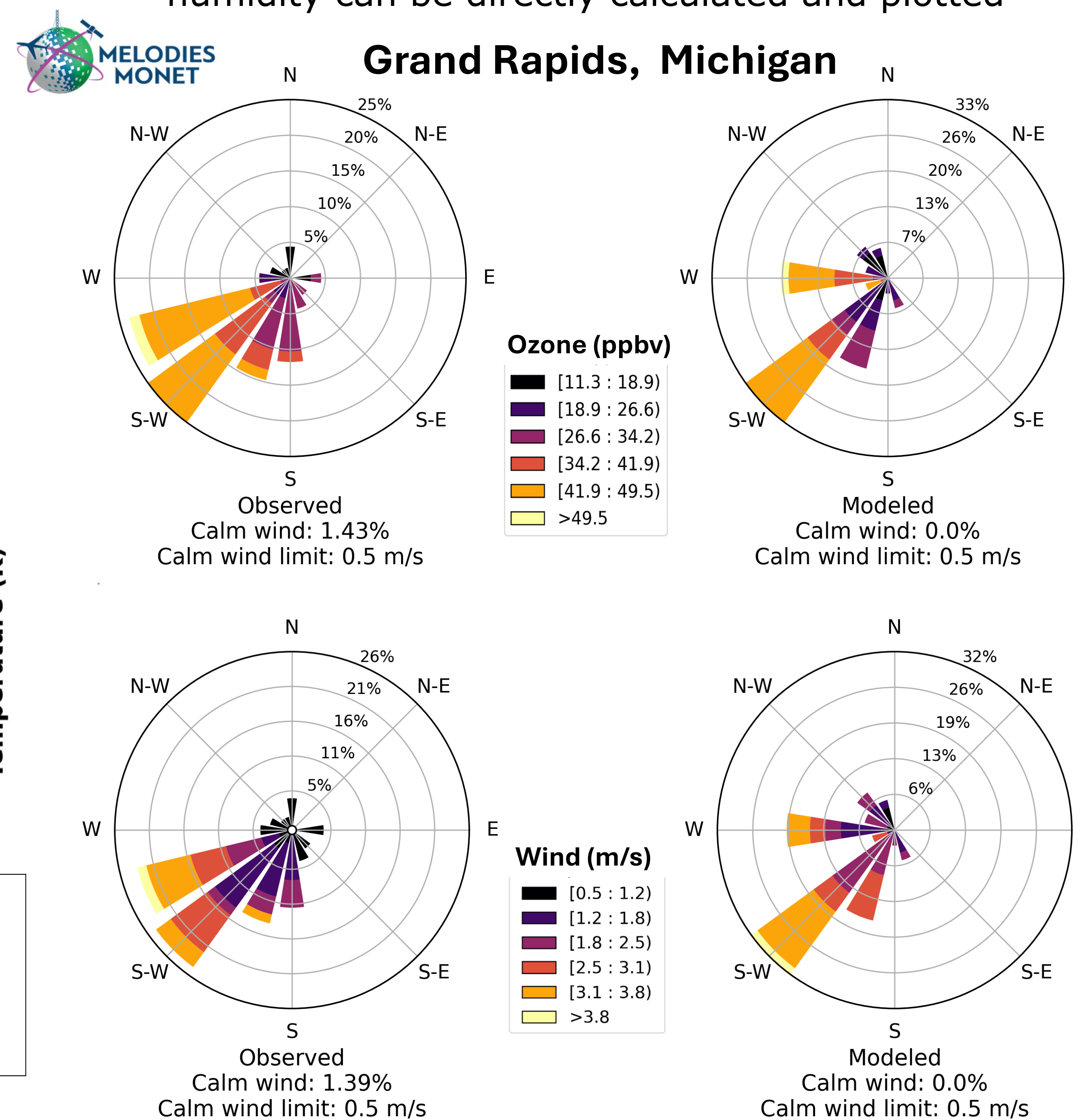


Fig. 3: Users can now create meteorological & pollution roses to analyze variables such as wind speed and ozone

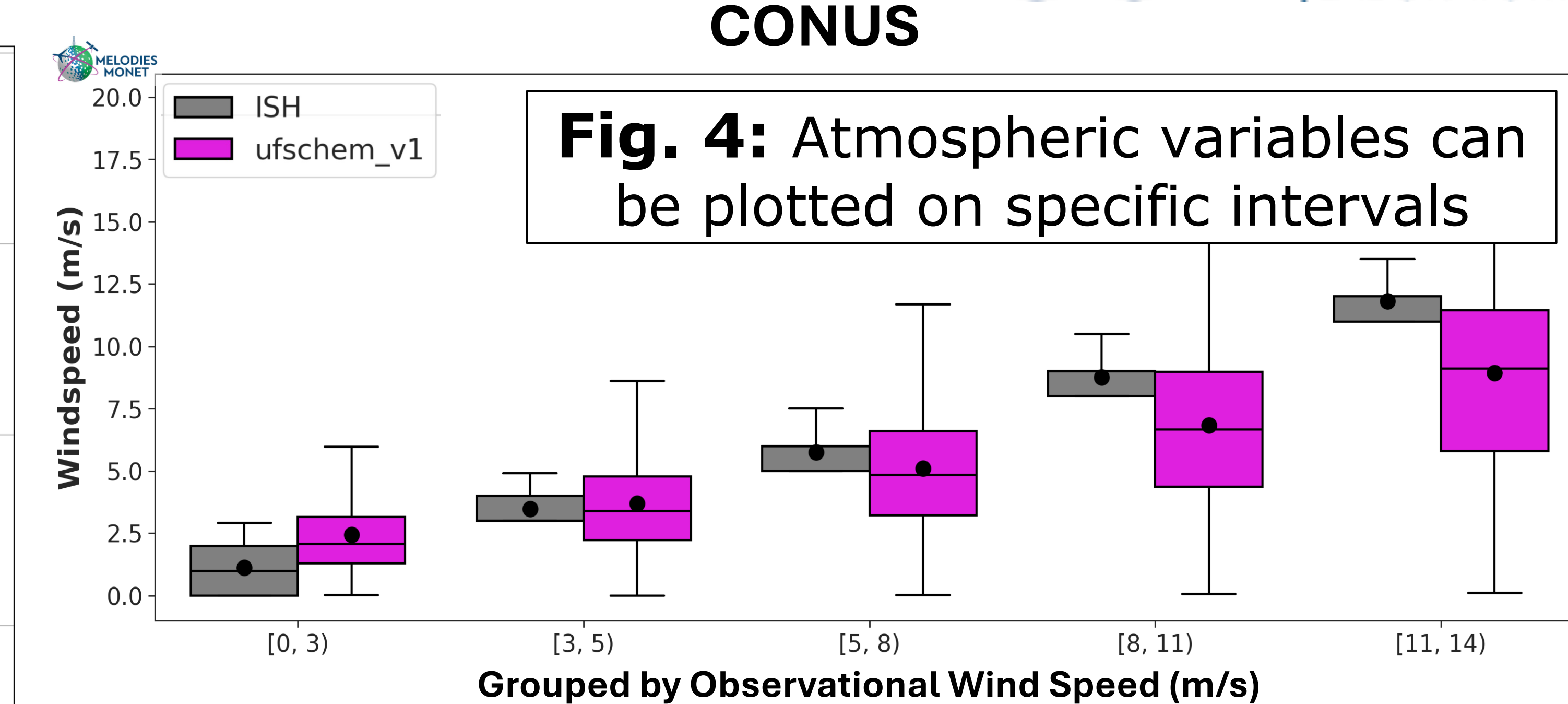


Fig. 4: Atmospheric variables can be plotted on specific intervals

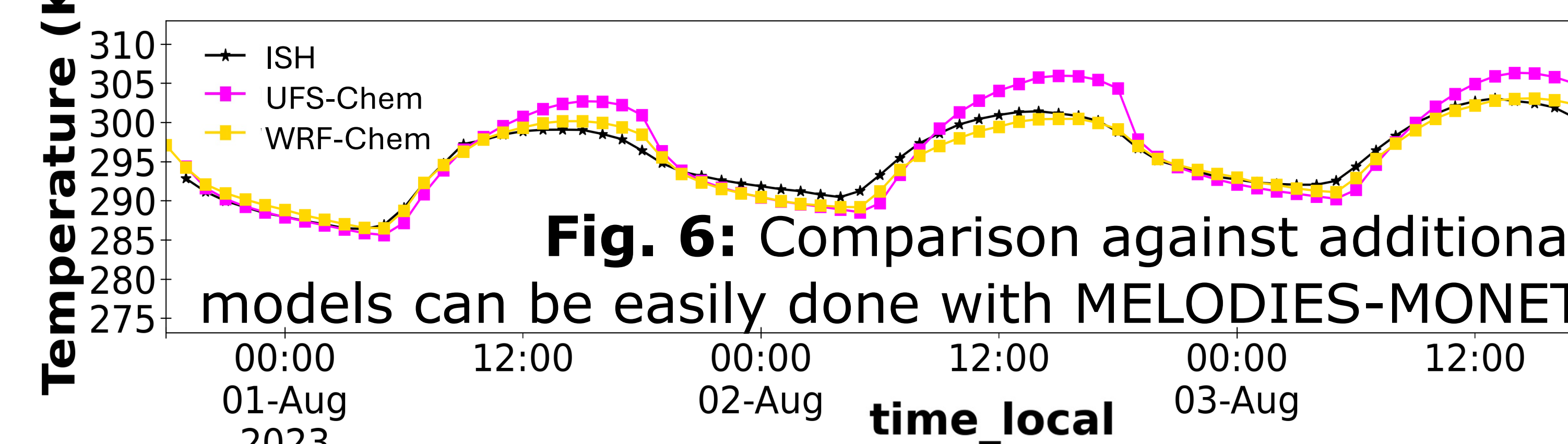
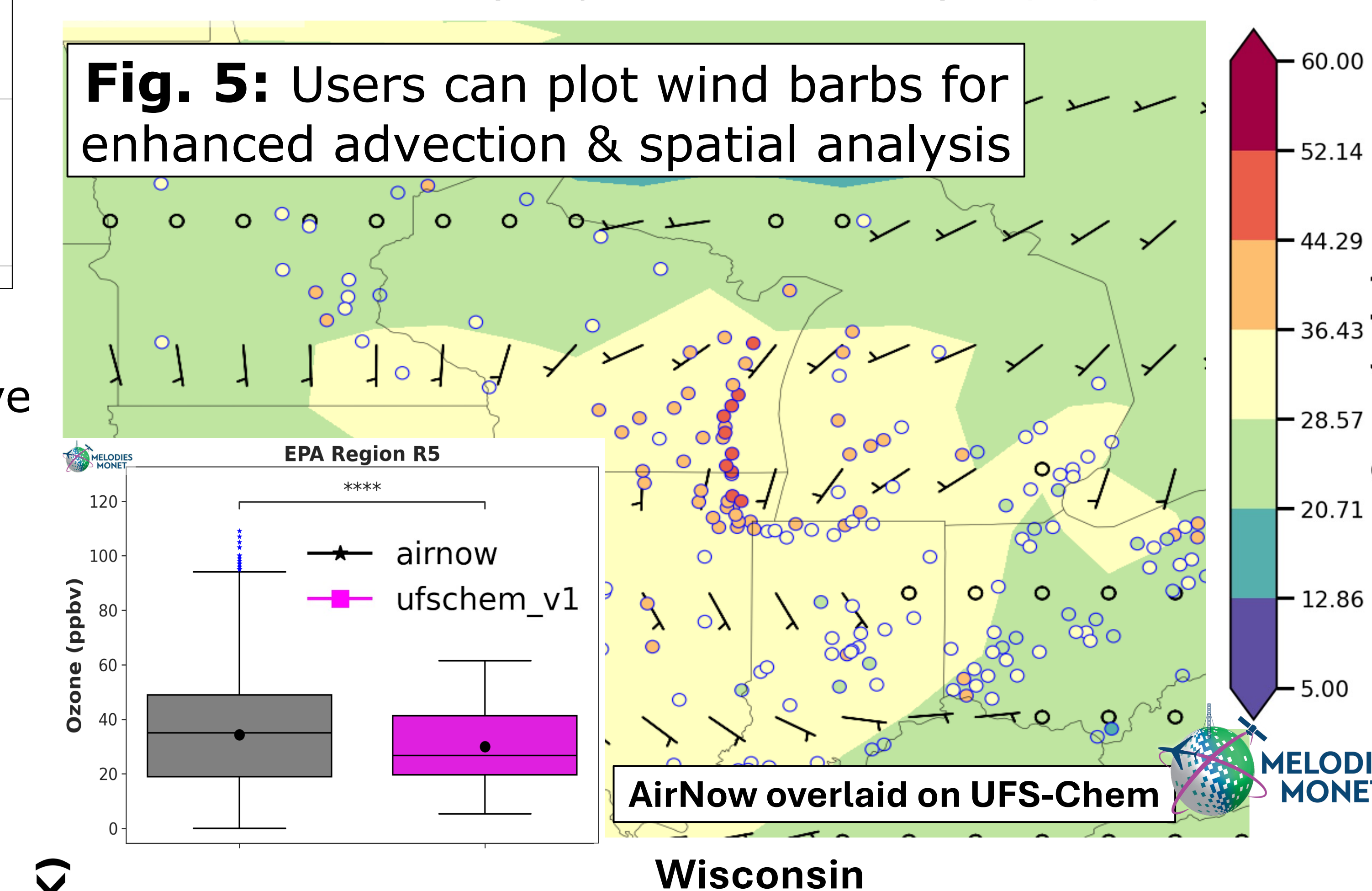


Fig. 6: Comparison against additional models can be easily done with MELODIES-MONET

iii) Implications

- MELODIES-MONET enables users to perform quick model evaluations to determine the role meteorology could have in driving biases within their atmospheric chemistry models

iv) Acknowledgements

I want to thank & recognize Dr. Rebecca Schwantes for her exceptional mentorship throughout my research. I also want to recognize the guidance of Drs. Maggie Bruckner, Pablo Lichtig, & Brian McDonald. This research was supported and made possible by the Ernest F. Hollings Scholarship Program (15 U.S. Code § 1542).